PubTable-AI: An AI copilot for data curation from publication tables

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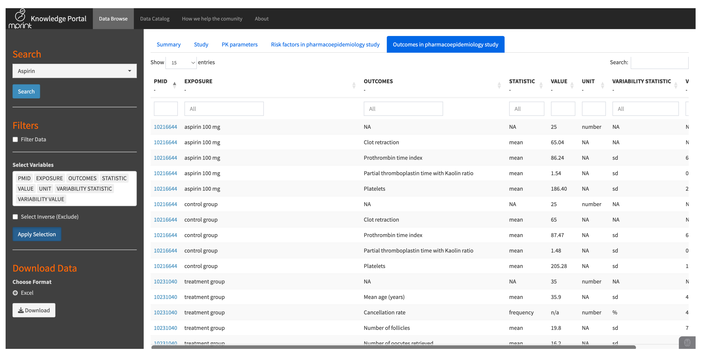
Maternal Pediatric Precision in Therapeutics Hub

- The MPRINT Hub serves as a national resource for expertise in maternal and pediatric therapeutics to conduct and foster therapeutics-focused research in obstetrics, lactation, and pediatrics while enhancing the inclusion of people with disabilities.
- MPRINT provides a knowledgebase of data and resources relating to maternal and pediatric therapeutics, including **pharmacokinetics, pharmacodynamics, pharmacogenetics, pediatric ontogeny** and the **physiological changes** that occur with and around pregnancy and lactation.

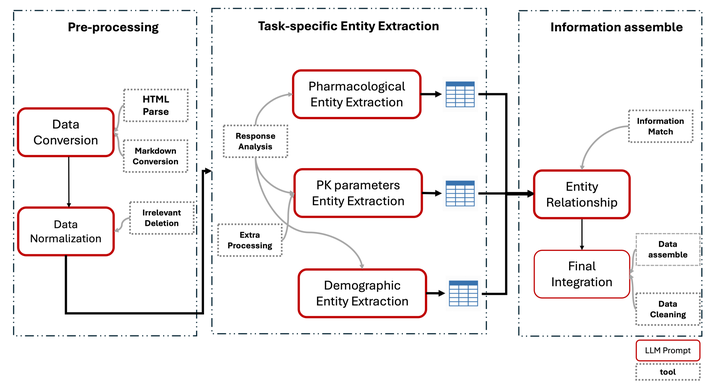
(a) MPRINT-KP **Silver:** 758,560 pharmacology research papers in maternal and pediatric populations by the BioBERT model.



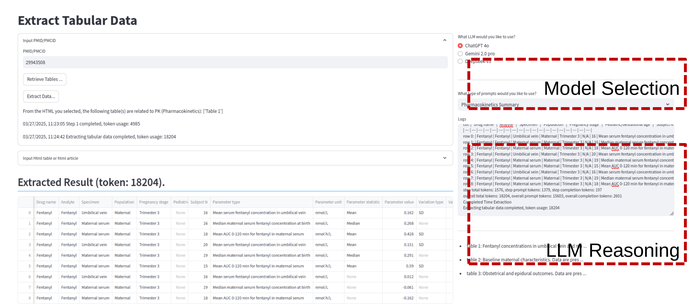
(b) MPRINT-KP **Gold**: 3947 pharmacology research papers in maternal and pediatric research curated by experts.

PubTable-AI

- **Task Weaver: Decomposed Prompt Workflow**



- **User-friendly UI & Automatic Workflow**



Primary Function and Features

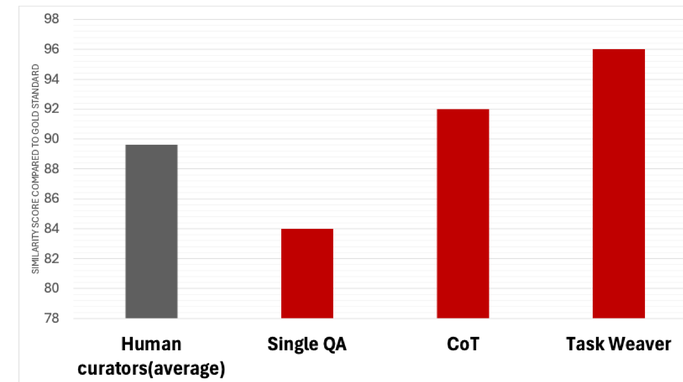
- Full-text automatic retrieve
- Automatic/Hyper table selection
- Multiple Large Language Model
- Reasoning for correction

Conclusion

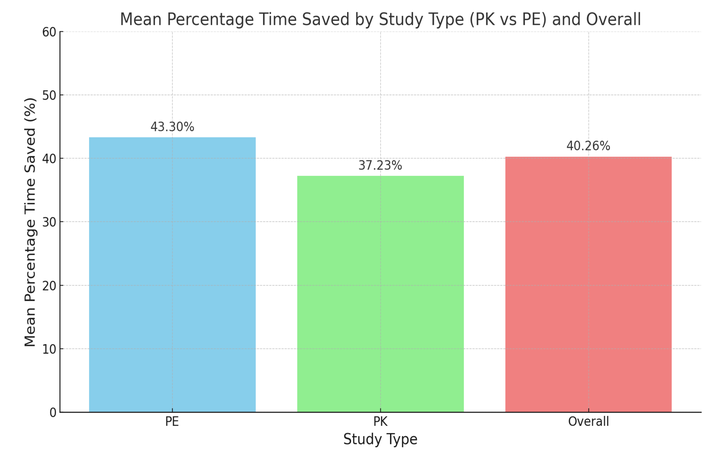
PubTable-AI is our recent attempt to enhance data curation tasks, providing biocurators with the efficiency and reliability of extracting data from scientific literature tables.

Superior Performance

- **Prompt Benchmark Performance (using GPT4o)**



- **Efficiency Performance**



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